

Phase-Controlled Multi-Ring Inertial Torque Modulation System for Enhanced Driveline Performance

Version 1.0 — Concept Whitepaper

Status: Conceptual / simulation-validated (no hardware prototype)

Abstract

Electric and hybrid powertrains often cannot deliver short, high-torque pulses without oversizing the battery and traction motor or accepting accelerated battery degradation. This paper introduces the **Phase-Controlled Multi-Ring Inertial Torque Modulation System** — an electromechanical buffer that uses multiple **mechanically independent** coaxial inertial rotors, each driven by its own electromagnetic motor-generator and supported on magnetic bearings, to store, shape, and release rotational energy on millisecond-to-second timescales.

A real-time controller coordinates phase, amplitude, and frequency of each rotor's acceleration profile so that reaction torques combine constructively at the output during peak demand and can be shaped to reduce driveline ripple in steady operation. **Illustrative** lumped-parameter simulations of a representative three-rotor auxiliary unit indicate **brief, energy-limited** peak torque increases **on the order of 35–50%** above a fixed motor baseline for **sub-second pulses** only—bounded by stored kinetic energy, electrical drive ratings, and simplified model assumptions (§4.4); **sustained torque multiplication is not claimed**.

The concept extends established flywheel/KERS principles with software-defined, multi-actuator torque shaping. It remains at an **early maturity stage** (conceptual; no hardware prototype): round-trip efficiency (**75–85% is an optimistic, unvalidated target** for a multi-rotor path with gears and multiple inverters), packaging, cost, and safety containment require hardware validation before automotive deployment. Near-term value is most plausible in **commercial stop-start, industrial peak-torque, and grid-frequency** applications rather than mass-market passenger EVs without further development.

Keywords: Inertial torque modulation, independent multi-rotor flywheel, hybrid powertrain, magnetic bearings, software-defined driveline, torque shaping, kinetic energy buffer.

1. Introduction

Modern electric vehicles (EVs) and hybrid electric vehicles (HEVs) face a recurring trade-off: components sized for energy efficiency and battery life are often undersized for brief torque spikes during hard acceleration, lane changes on grades, or repeated stop-start duty cycles. Mitigations include larger battery packs, higher-power inverters, supercapacitor buffers, or mechanical flywheel Kinetic Energy Recovery Systems (KERS). Each option involves compromises in cost, mass, complexity, efficiency, or control fidelity.

Flywheel KERS has demonstrated value in motorsport (Flybrid, Williams/GKN Gyrodrive) and selected commercial programs. Most deployments use **one** high-speed rotor with limited ability to sculpt the time-domain torque waveform at the wheel. Supercapacitors excel at electrical pulse power but store less energy per unit mass than flywheels for sustained bursts.

This paper proposes a **Phase-Controlled Multi-Ring Inertial Torque Modulation System**: several independent inertial rotors on a common axis, each with dedicated motor–generator windings and magnetic bearing support. Torque from each rotor is summed onto a central output path. Software coordinates per-rotor phase so pulses add when needed and can be arranged to limit net gyroscopic moment (via counter-rotating pairs).

The document specifies a reference mechanical architecture, states explicit energy and power limits, reports illustrative simulation metrics, and outlines a prototype roadmap.

2. System Description

The system is an electromechanical subsystem integrated between a primary energy source (battery + traction motor, or engine–generator) and the driveline. It acts as an **active inertial buffer** and **torque shaper**, not a replacement for the primary power source.

2.1 Reference Architecture

The recommended reference design uses **independent coaxial inertial rotors** (nested rings) whose rotational speeds are **not** kinematically locked to one another. Each rotor is supported on its own magnetic bearing set and driven by a dedicated motor–generator (MG) stator. Reaction torque from each MG stator is transmitted to a **torque-summing stage** that couples to the central output shaft.

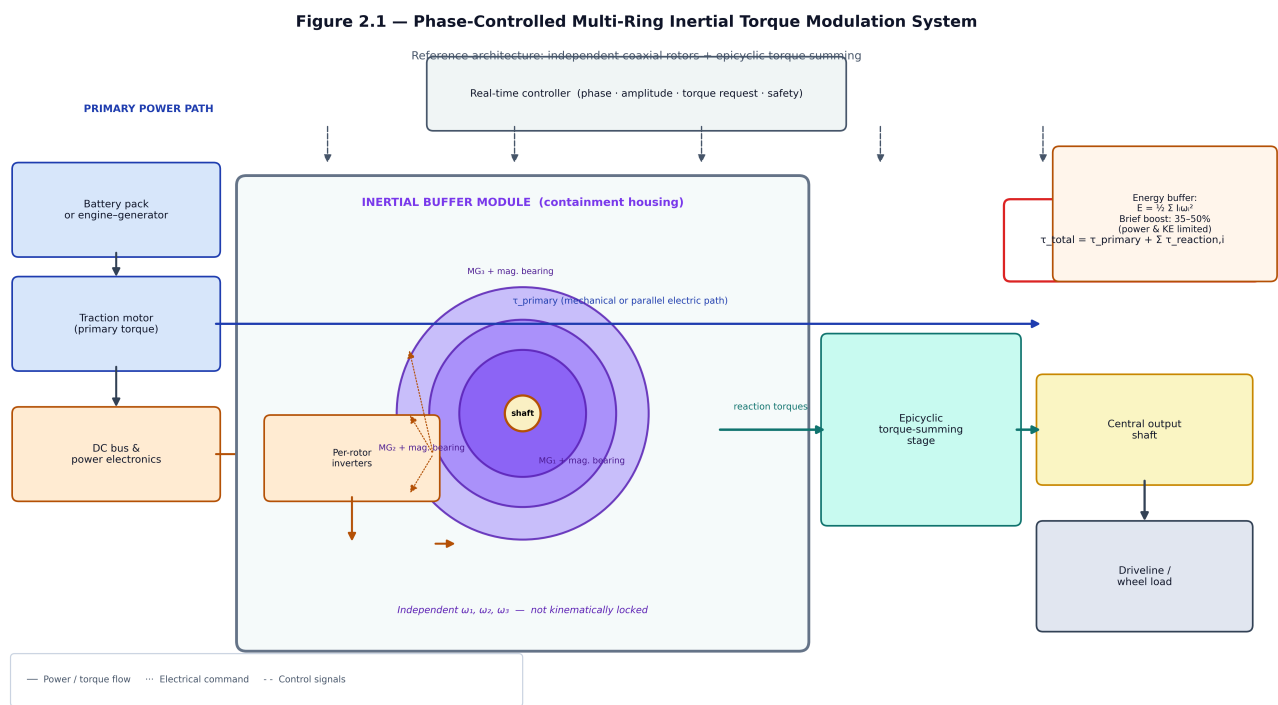


Figure 2.1 — Phase-controlled multi-ring inertial torque modulation system (reference architecture).

Torque-summing options (reference vs. alternatives):

Option	Description	Reference?	Notes
A. Epicyclic summing	Each rotor MG stator grounded to a carrier; planet/carrier torque sums to	Primary reference	Mature topology; predictable torque addition; adds gear losses (~1–3%)

Option	Description	Reference?	Notes
	output		per stage)
B. Direct MG reaction	Stator of each in-wheel or co-axial MG bolted to summing housing	Alternative	Fewer gears; requires stiff housing and thermal design
C. Magnetic torque coupling	Non-contact torque transmission between rotor and output	Research option	No contact wear; air-gap control and fault tolerance are challenging

Reference selection: Option **A (epicyclic summing)** is the default reference for analysis and prototyping because it is well understood in hybrid transmissions and allows independent rotor speeds while providing a single mechanical output.

Primary electrical power flows: battery → DC bus → per-rotor inverters → MG windings. The controller commands torque and speed trajectories; encoders close the loop on each rotor.

2.2 Key Components

Independent inertial rotors — Multiple coaxial rings, each with moment of inertia I_i , speed $\omega_i(t)$, and no rigid kinematic tie to other rotors.

Electromagnetic drive units — One MG per rotor; bandwidth target >200 Hz torque command (subject to inverter and control loop limits).

Magnetic bearings — Active or hybrid magnetic bearings per rotor to minimize friction and allow differential speeds.

Torque-summing stage — Epicyclic or equivalent mechanism aggregating reaction torques (see §2.1).

Sensors and controller — Per-rotor position/speed, bus voltage/current, vehicle CAN inputs; embedded real-time optimizer for A_i, Ω_i, ϕ_i .

2.3 Coaxial Rotor Cross-Section (Conceptual)

The buffer module contains nested independent rotors (see Figure 2.1, center). Each ring has its own magnetic bearing set and motor-generator; reaction torques route to the epicyclic summing stage and central output shaft.

2.4 Operational Modes

Mode	Objective	Typical behavior
Torque boost	Short acceleration pulse	Rotors discharge stored KE; phases aligned for constructive reaction torque
Energy recovery	Braking / downhill	Rotors accelerate, absorbing driveline energy
Smoothing	Reduce ripple	Phases adjusted to cancel oscillatory components
Decoupling	Protect battery/motor	Primary source held near efficient point; buffer supplies transient deficit

2.5 Representative Sizing (Illustrative Auxiliary Unit)

The following **illustrative** sizing bracket is used for simulation and early BOM thinking—not a committed product specification.

Parameter	Value (illustrative)
Number of rotors	3 (expandable to 4–6)
Per-rotor inertia	0.10–0.15 kg·m ²
Operating speed (mean)	~7,600 rpm (800 rad/s)
Stored energy (total)	~0.12 MJ (118 kJ)
Per-rotor MG continuous rating	30–35 kW
Pack electrical rating (3 rotors)	~90–105 kW
Target peak boost (brief)	+35–50% vs. motor-only baseline
Boost duration (typical)	0.2–0.5 s (energy-limited)
Target round-trip efficiency	75–85% (to be validated; includes MG, bearings, gears, power electronics)
Envelope target (automotive aux.)	~40–60 L, 45–70 kg (competitive with single high-speed flywheel module)

3. Working Principle

3.1 Per-Rotor Torque

For rotor i , electromagnetic torque applied to the rotor:

$$\tau_{\text{coil},i} = I_i \cdot \alpha_i$$

Reaction torque on the stator/housing (and thus the summing stage):

$$\tau_{\text{reaction},i} = -\tau_{\text{coil},i} = -I_i \cdot \alpha_i$$

3.2 Total Output Torque

$$\tau_{\text{total}}(t) = \tau_{\text{primary}}(t) + \sum_{i=1}^n \tau_{\text{reaction},i}(t)$$

where τ_{primary} is torque from the main traction motor (or engine path) upstream or in parallel, depending on integration.

3.3 Phase-Controlled Modulation

Independent rotors allow separate velocity programs:

$$\omega_i(t) = \omega_{0,i} + A_i \sin(\Omega_i t + \phi_i)$$

$$\alpha_i(t) = A_i \Omega_i \cos(\Omega_i t + \phi_i)$$

The controller selects $\mathbf{A}_i$, Ω_i , and ϕ_i to meet a torque request while respecting:

- DC bus voltage and current limits
- Maximum ω_i and α_i
- Minimum stored energy $E_i = \frac{1}{2} I_i \omega_i^2$
- Thermal limits on windings and bearings

3.4 Counter-Rotating Pairs (Gyroscopic Management)

For vehicles sensitive to gyroscopic precession, rotor pairs can be operated with opposite mean angular momentum $\mathbf{H}_i = I_i \omega_i \mathbf{n}$, while still using phase modulation on each pair member for torque shaping. This does not eliminate gyroscopic effects under all maneuvers but can reduce net angular momentum compared to a single large flywheel.

3.5 Energy and Power Limits (Design Constraint)

Peak output torque cannot exceed what stored kinetic energy and electrical input can support. For a pulse of duration Δt and incremental torque $\Delta \tau$:

$$E_{\text{stored}} \gtrsim \int \Delta \tau \cdot \omega_{\text{out}} dt$$

Sustained torque multiplication (e.g., holding 8–10× baseline for seconds) is **not physically realistic** without very large inertia, speed, and electrical power—far beyond practical vehicle packaging. The system is intended for **brief, shaped pulses** and driveline quality improvement, not continuous overpowering of the traction motor.

4. Simulation Results

4.1 Model Assumptions

A lumped-parameter model was implemented in Python (NumPy) with:

- Three independent rotors ($I = [0.12, 0.15, 0.10]$ kg·m²)
- Mean speed 800 rad/s per rotor
- Sinusoidal speed modulation with $\Omega = 18$ rad/s and 120° phase separation
- Fixed primary torque baseline $\tau_{\text{primary}} = 180$ N·m
- Simplified neglect of epicyclic losses, bearing drag, and inverter limits (conservative torque capability; efficiency not yet back-calculated)

Full code is provided in the **Appendix**.

4.2 Results Summary

Metric	Value
Baseline torque (motor only)	180 N·m
Peak combined torque	242.8 N·m
Peak torque increase	+34.9% (brief)

Metric	Value
Total stored kinetic energy (mean speed)	0.118 MJ
Approx. peak instantaneous electrical demand	~305 kW*

*Instantaneous sum of $|\tau_i \omega_i|$ across rotors; average power over a duty cycle is lower. Production design would cap commands to per-inverter ratings (~30–35 kW continuous per rotor).

4.3 Interpretation

Figure 4.1 — Torque augmentation

Primary torque (flat 180 N·m) plus summed reaction torque from three phase-shifted rotors yields a peak near **243 N·m**. Alignment of phases produces a narrow constructive peak; misalignment reduces peak but can smooth ripple.

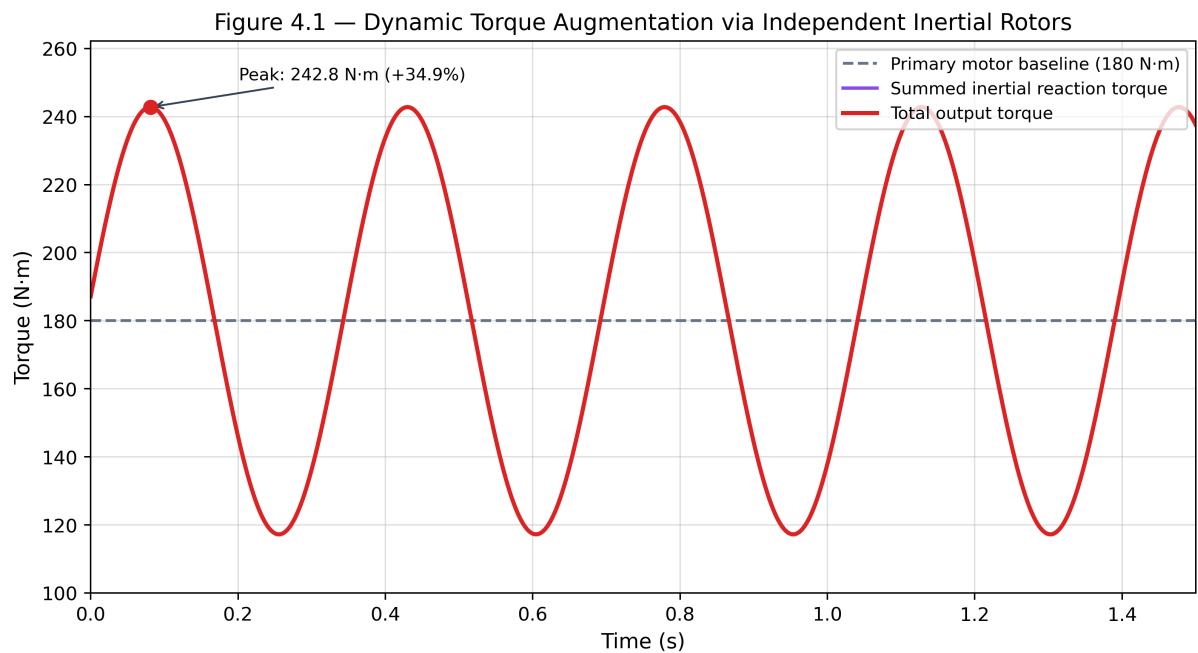
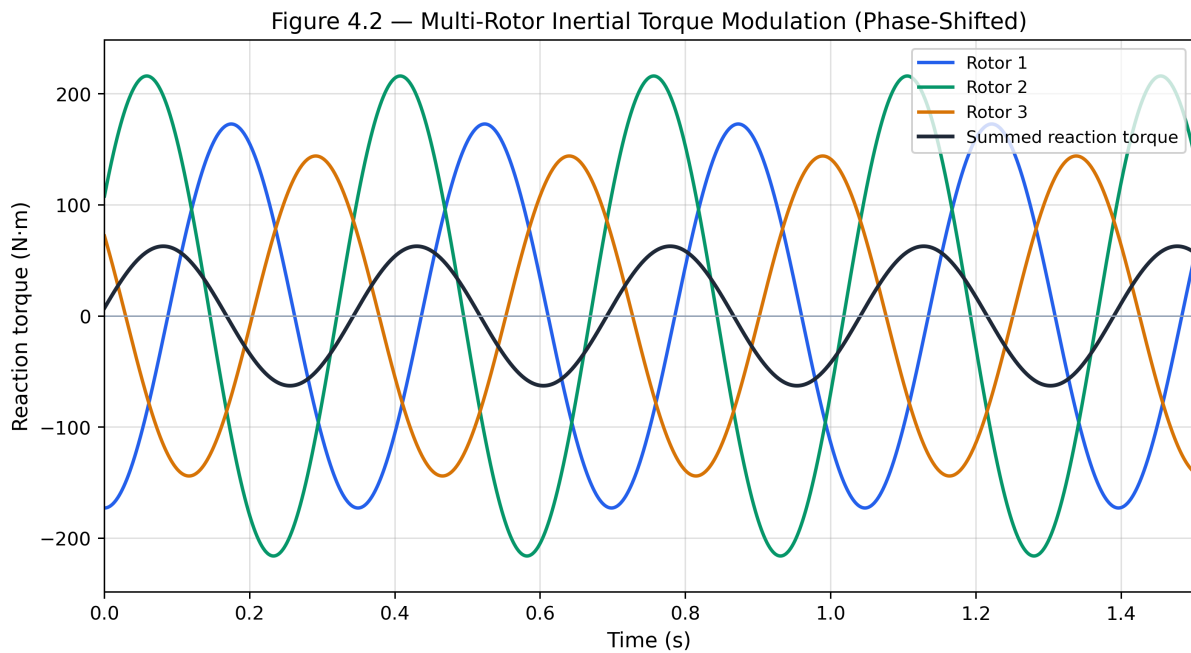


Figure 4.2 — Per-rotor reaction torque

Each rotor contributes a sinusoidal reaction component; superposition creates the combined peak.



Figures generated by `whitepaper_plots.py` (Python / Matplotlib). A combined two-panel overview is also available as `whitepaper_fig_combined_overview.png`.

4.4 Limitations of Current Simulation

- No multi-body/epicyclic dynamics or gear mesh stiffness
- No inverter saturation or thermal derating
- No efficiency map or round-trip energy accounting
- No vehicle longitudinal model (mass, tire, grade)

Future simulations will incorporate detailed loss models (copper, iron, bearing, and epicyclic) to produce realistic efficiency maps.

Next modeling step: co-simulation with vehicle dynamics and loss models; cap peak power to per-inverter ratings; predict fuel/energy benefit on standard drive cycles (WLTC, CARB bus cycle).

5. Comparison with Existing Technologies

Table 5.1 — Qualitative comparison

Feature	Battery + supercapacitors	Single flywheel KERS	Proposed multi-rotor system
Torque response	Fast (electrical)	Very fast	Very fast (electrical + inertia)
Waveform shaping	Limited	Low	High (multi-actuator phase control)
Peak torque boost	Moderate	High (short)	High (brief , energy-limited)
Decoupling primary source	Moderate	Good	Good (buffer absorbs transients)

Feature	Battery + supercapacitors	Single flywheel KERS	Proposed multi-rotor system
Gyroscopic effects	None	Significant	Reduced with counter-rotating pairs
Round-trip efficiency	Good (cap)	High (flywheel)	75–85% target (TBD)
Power density	High	Very high	Very high
Control flexibility	High	Moderate	High
Mechanical complexity	Low	High	High
Technology maturity	High	Medium–high	Low (concept)

Strengths: Real-time torque shaping; software-updatable control; potential to downsize primary battery/motor for same perceived performance.

Weaknesses: Multiple high-speed rotors, bearings, and inverters; containment and cost; must prove efficiency and reliability vs. simpler single flywheel or cap bank.

6. Potential Applications

6.1 Automotive & mobility — Commercial trucks and buses (stop–start), performance hybrids, niche EV torque enhancement. Mass-market passenger EV adoption is **uncertain** until cost and packaging match supercapacitor or motor upsizing.

6.2 Industrial & machinery — Cranes, presses, material handling with repetitive peak loads.

6.3 Renewable & grid — Frequency regulation, wind/solar smoothing (competes with established grid flywheels).

6.4 Defense & aerospace — Burst power for ground support; drone launch systems (subject to weight limits).

Modular scaling: auxiliary units ~20–50 kW (light commercial) to >150 kW (heavy duty), subject to containment and bearing development.

7. Safety, Efficiency, and Integration

Safety: High-speed rotors require burst containment per industry practice (e.g., SAE practices for energy-storage devices), rotor burst sensors, and safe shutdown on bearing or inverter fault.

Efficiency: Target **75–85%** round-trip (charge/discharge cycle) at module level (**TBD** — likely optimistic until loss models and hardware tests account for copper, iron, bearing, epicyclic, and multi-inverter overhead); losses must be measured on hardware.

Integration: Parallel connection to driveline (torque additive) or series hybrid node; requires OEM collaboration for packaging, NVH, and fault diagnostics (ISO 26262 ASIL assignment TBD).

8. Future Work and Roadmap

8.1 Key Challenges

- Independent coaxial rotor mechanical design and manufacturing tolerances
- **Simulation realism** — extend beyond lumped-parameter models; future work will incorporate detailed loss models (copper, iron, bearing, and epicyclic) to produce realistic efficiency maps (see §4.4)
- Torque-summing stage losses and NVH
- Real-time multi-rotor control stability and phase optimization
- Thermal management of concentrated windings
- Cost vs. supercapacitor / motor resize

8.2 Development Roadmap

Phase	Timeline (indicative)	Deliverables
1 — Modeling	0–6 months	MBD with epicyclic dynamics; efficiency map; drive-cycle energy benefit
2 — Single-rotor lab	6–12 months	One MG + magnetic bearing + dynamometer; validate $\tau = I\alpha$ and losses
3 — Dual/triple rotor	12–24 months	Phase control; summing stage; gyroscopic pair demonstration
4 — Vehicle demo	24–36 months	Bus or truck pilot; durability and containment test

8.3 Intellectual Property

Prior art exists in flywheel hybrids and multi-motor drivelines; novelty claims should focus on **independent coaxial multi-rotor phase coordination for driveline torque shaping** and specific summing architectures. Formal prior-art search and patent filing are recommended before public disclosure beyond this document.

9. Conclusion

The Phase-Controlled Multi-Ring Inertial Torque Modulation System is a **credible early-stage concept** that combines proven flywheel buffering with multi-actuator, software-coordinated torque shaping. The revised architecture—**independent coaxial rotors** with an explicit **torque-summing stage**—resolves prior kinematic ambiguities and aligns with the control model.

Illustrative simulation supports **brief peak torque increases of roughly 35%** above a fixed motor baseline for a representative three-rotor unit, within stated energy and power constraints. The concept is **not** validated in hardware; efficiency, cost, safety, and competitive positioning against supercapacitors and single flywheels remain open.

Recommended next steps: high-fidelity MBD, single-rotor lab prototype, and duty-cycle-based comparison on commercial-vehicle cycles where flywheel buffers have historically shown the strongest business case.

Appendix A — Simulation Code

```
"""
Illustrative multi-rotor torque modulation (lumped parameter).
Run: py whitepaper_sim.py
"""

import numpy as np

n_rotors = 3
I = np.array([0.12, 0.15, 0.10])      # kg·m²
omega0 = np.array([800.0, 800.0, 800.0]) # rad/s
A = np.array([80.0, 80.0, 80.0])
Omega = np.array([18.0, 18.0, 18.0])
phi = np.array([0.0, 2 * np.pi / 3, 4 * np.pi / 3])

t = np.linspace(0, 1.5, 3000)
tau_primary = 180.0
tau_reaction = np.zeros_like(t)

for i in range(n_rotors):
    alpha = A[i] * Omega[i] * np.cos(Omega[i] * t + phi[i])
    tau_reaction += -I[i] * alpha

tau_total = tau_primary + tau_reaction

print("Peak total torque (N·m):", tau_total.max())
print("Boost (%):", 100 * (tau_total.max() - tau_primary) / tau_primary)
print("Stored energy (MJ):", 0.5 * np.sum(I * omega0**2) / 1e6)
```

Appendix B — Nomenclature

Symbol	Meaning
I_i	Moment of inertia of rotor i
ω_i	Angular velocity of rotor i
α_i	Angular acceleration of rotor i
$\tau_{\text{coil},i}$	Electromagnetic torque on rotor i
$\tau_{\text{reaction},i}$	Reaction torque on stator/summing stage
A_i, Ω_i, ϕ_i	Modulation amplitude, frequency, phase
E_i	Kinetic energy stored in rotor i

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Document prepared for technical review and partner discussions.

Related files (same folder):

File	Purpose
whitepaper_sim.py	Print simulation metrics to console
whitepaper_plots.py	Regenerate Figure 4.1 and 4.2 PNGs
whitepaper_system_diagram.py	Regenerate Figure 2.1 system architecture PNG
whitepaper_to_pdf.py	Build Whitepaper-Complete.pdf
whitepaper_fig2_1_system_architecture.png	Figure 2.1 — full system
whitepaper_fig4_1_torque_augmentation.png	Figure 4.1
whitepaper_fig4_2_per_rotor_torque.png	Figure 4.2

File	Purpose
whitepaper_fig_combined_overview.png	Optional two-panel summary slide
Whitepaper-Complete.pdf	PDF export